
Sampling Hidden Layers Like the Output Softmax: A Controlled Study of Intermediate-Layer Activation Sampling

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Abstract

The final layer of a classifier is stochastic: we form a softmax distribution over the classes and sample from it. We ask what happens when we do the same thing one layer earlier — treat an intermediate layer’s activations as a distribution and draw a sample, instead of passing the deterministic activations forward. Hidden-layer stochasticity is the shared mechanism behind variational autoencoders, Gumbel-Softmax latents, stochastic neurons, dropout, and the information bottleneck, but each of these studies it for a purpose or at a fixed place; no prior work isolates intermediate-layer activation sampling as the variable of study, and the analogy to sampling the output softmax is rarely made explicit. We run that controlled study with a single drop-in sampling layer, holding architecture, data, and optimizer fixed while varying only the activation operator across seven choices: deterministic, dropout, additive Gaussian noise, continuous Gaussian reparameterization, Bernoulli straight-through, and categorical Gumbel-Softmax in soft and hard forms. Across MNIST (MLP) and CIFAR-10 (CNN), 3 seeds and 117 trained models, we measure accuracy, NLL, calibration, generalization gap, gradient variance, MC-averaging, and sweeps over temperature, layer position, and input noise. We find that intermediate-layer sampling is a distribution-type-dependent regularizer. Continuous Gaussian sampling is essentially free (97.7% vs 97.8% on MNIST) and on CIFAR-10 beats the deterministic baseline on accuracy (75.5% vs 75.0%), NLL (0.78 vs 0.94), and calibration (ECE 0.074 vs 0.112). The literal softmax analogue is the opposite: categorical sampling costs 8–10 points on MNIST and collapses to chance (10%) on CIFAR-10, and its gradient variance is the highest of all operators, exactly tracking the accuracy ranking. MC-averaging recovers most of the single-sample cost, mirroring how averaging softmax samples approximates the expectation. Sampling a hidden layer like the output softmax is therefore possible and even beneficial in its gentle continuous form, but the literal categorical form is a severe information bottleneck whose cost scales with task difficulty.

1 Introduction

The final layer of a classifier is stochastic. The softmax defines a categorical distribution over classes, and at inference we either sample from it or take its argmax ; at training time the same distribution is what the cross-entropy loss scores. The hidden layers, by contrast, are deterministic: a ReLU activation passes its exact value forward, unchanged from pass to pass. This asymmetry is so familiar that it is rarely questioned. We ask the sharp version of the question it invites: **what if we apply the output-layer operation at an intermediate layer?** That is, form a probability distribution over a hidden layer’s activations and draw a *sample* from it, then pass that sample forward in place of the deterministic activations. The literal form of this idea treats a hidden layer exactly like

the output layer — a softmax over its units gives a categorical distribution, and we sample one unit, just as we sample one class.

Hidden-layer stochasticity is not new; it is the shared mechanism behind a remarkable range of methods. Variational autoencoders sample a continuous latent through the reparameterization trick [Kingma and Welling, 2014, Rezende et al., 2014]; Gumbel-Softmax and the Concrete distribution sample discrete latents differentiably [Jang et al., 2017, Maddison et al., 2017]; stochastic neurons pass binary samples with straight-through gradients [Bengio et al., 2013]; dropout injects multiplicative Bernoulli noise that doubles as a Bayesian approximation [Srivastava et al., 2014, Gal and Ghahramani, 2016]; and the deep variational information bottleneck deliberately makes a representation stochastic to limit the information it carries [Alemi et al., 2017, Tishby and Zaslavsky, 2015]. In every case the stochasticity is introduced *for a purpose* — generative modeling, discrete latents, regularization, compression — and *at a fixed place* chosen for that purpose. A practitioner who simply wants to know the cost (in accuracy) and the benefit (in calibration, generalization, and robustness) of sampling a hidden layer, and where in the network to place it, has no controlled answer to point to.

The gap. No prior work isolates intermediate-layer activation sampling as the variable of study itself. The methods above each entangle the sampling operation with an objective, an architecture, and a placement, so the effect of the sampling *per se* is never measured against an otherwise-identical deterministic network. The softmax-sampling analogy — treat a hidden layer exactly as the output layer, softmax over its units to a categorical distribution and sample one — is almost never made explicit, and its consequences are essentially unstudied. We close this gap with a single drop-in module and a controlled comparison.

Our approach. We build one `SAMPLINGLAYER` that, placed after a hidden ReLU, turns the incoming activations into a distribution and draws a sample. Holding architecture, data, and optimizer fixed, we compare seven activation operators: a DETERMINISTIC reference; two noise baselines (DROPOUT and additive GAUSSIAN-NOISE); and the three faces of the idea the literature identifies — continuous GAUSSIAN-REPARAM (VAE-style), binary Bernoulli straight-through (BERNOULLI-ST, the Bengio stochastic neuron), and categorical Gumbel-Softmax in its soft (GUMBEL-SOFTMAX) and hard one-hot (GUMBEL-ST) forms, the direct softmax analogue that collapses the layer to a single sampled unit. We evaluate on MNIST (MLP) and CIFAR-10 (CNN) across 3 seeds — 117 trained models in total — measuring accuracy, NLL, ECE calibration, the train–test generalization gap, gradient variance at the sampling layer, and single-sample versus MC-averaged inference, with sweeps over temperature/noise scale, layer position, and input-noise robustness.

What we find. Three results stand out. First, **the distribution type dominates everything**. Continuous GAUSSIAN-REPARAM sampling is essentially free — 97.7% vs 97.8% deterministic accuracy on MNIST (paired t -test $p = 0.83$, not significant) — and on the harder CIFAR-10 it *beats* the deterministic reference, cutting ECE from 0.112 to 0.074 and NLL from 0.94 to 0.78 at equal-or-better accuracy. The literal categorical analogue does the opposite: it is the most expensive operator on MNIST (−10.4 points for GUMBEL-ST, $p = 0.048$) and *collapses to chance* (10%) on CIFAR-10. Second, **sampling regularizes**. Every stochastic variant shrinks the train–test gap and most lower ECE — the benefit the literature predicts — so the gentle variants trade a tiny (or zero) accuracy cost for better-behaved probabilities. Third, **MC-averaging recovers the single-sample cost**, exactly as averaging softmax samples approximates the class expectation: dropout climbs from 96.6% to 97.9% as the number of test samples K grows from 1 to 50. Underlying all three, the gradient-variance ranking of the operators (lowest for Bernoulli straight-through, highest for hard categorical) tracks the accuracy cost almost exactly, giving a mechanistic explanation for why categorical sampling is so hard.

Contributions.

- We propose `SAMPLINGLAYER`, a single drop-in module that turns any hidden layer into a distribution and samples from it, exposing seven activation operators — a deterministic reference, two noise baselines, and the continuous, binary, and categorical faces of activation sampling — behind one interface.
- We conduct a controlled comparison that isolates intermediate-layer activation sampling as the variable of study, holding architecture, data, and optimizer fixed across MNIST and CIFAR-

10, 3 seeds, and 117 models, with accuracy, NLL, ECE, generalization gap, gradient variance, MC-averaging, and sweeps over temperature, layer position, and input noise.

- We show that continuous GAUSSIAN-REPARAM sampling is free and even beneficial (-0.03 pts on MNIST; $+0.5$ pts accuracy and 0.74 vs 0.94 NLL on CIFAR-10) while the direct categorical softmax analogue is the most expensive variant and collapses to chance on CIFAR-10, making explicit that “sampling a hidden layer like the softmax” is distribution-type-dependent rather than a free drop-in.
- We explain the cost mechanistically: the gradient variance induced at the sampling layer ranks the operators in the same order as their accuracy cost, identifying high-variance gradient estimation as the bottleneck for categorical sampling.

The rest of the paper is organized as follows. Section 2 places the seven operators within prior work on stochastic representations; Section 3 details the SAMPLINGLAYER and experimental protocol; Section 4 reports the core comparison, sweeps, and the CIFAR-10 confirmation; and Section 5 and Section 6 interpret the findings and outline next steps.

2 Related Work

Stochastic neurons and stochastic activations. The most direct precedent for our question replaces a deterministic hidden activation with a sample. Bengio et al. [2013] draw a binary value $h_i \sim \text{Bernoulli}(\sigma(a_i))$ at each hidden unit and introduce the straight-through estimator to pass gradients through the non-differentiable sample; this is exactly our BERNOULLI-ST mode. Lee et al. [2019] push the idea to every layer with ProbAct, a stochastic activation function whose output is sampled from a per-element (mean, variance) distribution with a learnable variance, reporting generalization gains. These methods establish that hidden-layer sampling is trainable and can help, but they fix the operator (Bernoulli, or a single Gaussian-like form) rather than comparing distribution types. We place the Bernoulli stochastic neuron on the same axis as continuous and categorical sampling and measure where each one falls.

Reparameterization and differentiable sampling. Sampling at a hidden layer blocks gradients unless the sample is made differentiable. For continuous activations, the reparameterization trick [Kingma and Welling, 2014, Rezende et al., 2014] writes $z = \mu + \sigma \odot \epsilon$ with parameter-free noise ϵ , giving a low-variance pathwise gradient; this is the machinery behind our GAUSSIAN-REPARAM mode. For categorical distributions, Gumbel-Softmax and the equivalent Concrete distribution [Jang et al., 2017, Maddison et al., 2017] relax a one-hot sample of $\text{softmax}(x)$ through a temperature- τ continuous approximation, recovering a hard sample as $\tau \rightarrow 0$. This is the most direct match to our motivating question: sampling a hidden layer the way we already sample the output softmax. We use it for GUMBEL-SOFTMAX (soft, $\tau > 0$) and GUMBEL-ST (hard one-hot via straight-through). Our contribution is to put the continuous and categorical reparameterizations on the same axis and measure the gap: the Gaussian path costs essentially nothing, whereas the categorical path carries the highest gradient variance and the largest accuracy penalty.

Discrete representation learning. VQ-VAE [van den Oord et al., 2017] places a categorical codebook at an intermediate layer and trains it with a straight-through gradient, demonstrating that discrete hidden representations are learnable. Crucially, it warns that a soft relaxation of the codebook can be inverted or ignored by a sufficiently strong downstream stack, and that the discrete hidden code can collapse to a few entries. Our CIFAR-10 result instantiates exactly this caution: when the categorical sample collapses a hidden layer to roughly one active unit, the harder task cannot recover and GUMBEL-ST falls to chance. Where VQ-VAE engineers around collapse to make discrete codes useful, we hold the architecture fixed and report collapse as a measured consequence of categorical hidden sampling.

Stochasticity as regularization, Bayesian inference, and robustness. A large body of work injects noise into a network for reasons other than studying activation sampling per se. Dropout and MC-dropout [Srivastava et al., 2014, Gal and Ghahramani, 2016] multiply activations by a Bernoulli mask, the latter reinterpreting test-time sampling as approximate Bayesian inference; variational dropout [Kingma et al., 2015] learns per-unit noise via the local reparameterization trick. Bayes-by-backprop [Blundell et al., 2015] instead samples the *weights* to capture parameter uncertainty — a deliberate contrast, since we sample activations and leave weights deterministic. Noisy activation functions [Gülçehre et al., 2016] add noise to ease optimization of saturating nonlinearities, and the

deep variational information bottleneck [Alemi et al., 2017, Tishby and Zaslavsky, 2015] samples a stochastic latent to compress nuisance information. Stochastic activation pruning [Dhillon et al., 2018] samples a sparse activation mask for adversarial robustness, though those robustness claims were later weakened under stronger adversarial evaluation. We adopt DROPOUT and GAUSSIAN-NOISE as baselines from this lineage and quantify the regularization effect each operator produces, measuring calibration with the expected calibration error (ECE) of Guo et al. [2017] alongside accuracy and the generalization gap.

Our position. Across all of this work, stochasticity is introduced either for a *purpose* — generation, compression, uncertainty estimation, or robustness — or at a *fixed place* — a latent bottleneck, a codebook, or every layer simultaneously. None isolates intermediate-layer activation sampling as the single variable of study under one protocol that sweeps distribution type, sampling amount, layer position, and train-versus-test usage while holding everything else constant. That is exactly what we do. By placing one SAMPLINGLAYER after a hidden ReLU and swapping only its sampling distribution, we make the analogy to output-softmax sampling explicit and ask what it costs to apply the same operation one layer earlier.

3 Methodology

We study a single, controlled intervention: take the activations of one hidden layer, turn them into a probability distribution, and *sample* from that distribution instead of passing the activations forward deterministically. This is exactly the operation applied to the final softmax of a classifier, except we apply it to an intermediate layer. To isolate this intervention, we keep the backbone, optimizer, data pipeline, and evaluation protocol fixed and vary *only* the distribution type and sampling rule at one layer.

3.1 The SAMPLINGLAYER

A drop-in stochastic module. The SAMPLINGLAYER is a module placed immediately after a hidden ReLU. It receives the post-ReLU activations $x \in \mathbb{R}^d$ of that layer, interprets them as the parameters of a distribution, draws a sample h , and forwards h in place of x . The rest of the network is untouched. We implement seven modes, summarized in table 1: a deterministic reference, two baselines that inject noise without learning a distribution (DROPOUT, GAUSSIAN-NOISE), one continuous learned distribution (GAUSSIAN-REPARAM), and three discrete operators (BERNOULLI-ST, GUMBEL-SOFTMAX, GUMBEL-ST). The discrete trio spans the straight-through stochastic neuron of Bengio et al. [2013] and the two direct analogues of softmax sampling.

The categorical modes implement the idea literally. For GUMBEL-SOFTMAX and GUMBEL-ST we treat the d units of the hidden layer exactly as we treat the C class logits at the output: we form a categorical distribution $\text{softmax}(x)$ over the units and draw a Gumbel-Softmax sample, the same operation we apply to the final softmax, but now over hidden units rather than classes. GUMBEL-SOFTMAX returns the relaxed simplex vector; GUMBEL-ST returns a hard one-hot sample, so the layer collapses to a single active unit per forward pass. To keep the forwarded magnitude comparable to a dense ReLU activation, we rescale the one-hot simplex vector by the layer width d , so the active unit carries mass of order one rather than $1/d$. This is the most aggressive intervention we test: it replaces a d -dimensional representation with a one-of- d choice.

The continuous mode carries a built-in regularizer. For GAUSSIAN-REPARAM, each unit emits a per-unit mean $\mu(x)$ and standard deviation $\sigma(x)$, and we draw

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (1)$$

using the reparameterization trick [Kingma and Welling, 2014]. We add a small KL-to-prior penalty $\text{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I))$ with weight 10^{-3} , following the variational autoencoder and deep variational information bottleneck [Kingma and Welling, 2014, Alemi et al., 2017]. This term is the only architectural extra any mode receives; it keeps the learned per-unit variances from collapsing or exploding.

The discrete baselines and the stochastic neuron. BERNOULLI-ST samples each unit independently as a hard 0/1 Bernoulli draw and uses a straight-through estimator for the gradient,

$$h_i \sim \text{Bernoulli}(\sigma(x_i)), \quad (2)$$

Table 1: The seven SAMPLINGLAYER modes. Each is a drop-in operator placed after a hidden ReLU. The categorical modes (GUMBEL-SOFTMAX, GUMBEL-ST) are the direct analogues of sampling the final softmax, applied to hidden units; GUMBEL-ST rescales its one-hot output by the layer width d to match dense-activation magnitude. ST = straight-through; RT = reparameterization trick.

Mode	Distribution	Sample	Gradient estimator	Role
DETERMINISTIC	—	x	exact	reference
DROPOUT	Bernoulli mask	$\text{mask} \cdot x / (1-p)$	exact	baseline (MC-dropout)
GAUSSIAN-NOISE	$x + \sigma \mathcal{N}(0, I)$	$x + \sigma \epsilon$	RT	baseline (unlearned noise)
GAUSSIAN-REPARAM	$\mathcal{N}(\mu(x), \sigma(x)^2)$	$\mu + \sigma \odot \epsilon$	RT (+ KL)	continuous (VAE-style)
BERNOULLI-ST	Bernoulli($\sigma(x)$)	hard 0/1	ST	stochastic neuron
GUMBEL-SOFTMAX	softmax(x) over units	relaxed simplex	Gumbel-Softmax	softmax analogue (soft)
GUMBEL-ST	softmax(x) over units	one-hot ($\times d$)	ST-Gumbel	softmax analogue (hard)

i.e., the Bengio stochastic neuron [Bengio et al., 2013]. The Gumbel-Softmax sample is

$$y_i = \text{softmax}\left(\frac{\log \pi_i + g_i}{\tau}\right), \quad g_i \sim \text{Gumbel}(0, 1), \quad (3)$$

with $\pi = \text{softmax}(x)$ and temperature τ [Jang et al., 2017, Maddison et al., 2017]. GAUSSIAN-NOISE adds reparameterized noise $x + \sigma \mathcal{N}(0, I)$ without learning the distribution, and DROPOUT applies the usual Bernoulli mask with inverted scaling $\text{mask} \cdot x / (1-p)$ and an exact gradient [Srivastava et al., 2014, Gal and Ghahramani, 2016]. These two baselines let us separate *learned* distributional sampling from generic noise injection.

3.2 Backbones, training, and evaluation

Backbones. On MNIST we use an MLP $784 \rightarrow 256 \rightarrow 256 \rightarrow 10$ with ReLU activations and place the SAMPLINGLAYER after the first hidden layer (position 0) by default [LeCun et al., 1998]. On CIFAR-10 we use a small CNN (4 convolutional blocks followed by 2 fully connected layers) and place the SAMPLINGLAYER on the 256-dimensional fully connected features [Krizhevsky, 2009]. Both backbones are otherwise standard; the only architectural difference across runs is which mode the SAMPLINGLAYER uses.

Training. We train with Adam (learning rate 10^{-3}), batch size 128, for at most 15 epochs on MNIST and 20 on CIFAR-10. We carve a 10% validation split from the training set (deterministic per seed) and use it for early stopping (patience 5, best checkpoint restored) and for selecting knobs such as τ and σ ; the test set is touched only for final evaluation. Standardization statistics are computed on the train split alone, so there is no leakage. We repeat every configuration over seeds $\{0, 1, 2\}$.

Metrics. We report test accuracy, NLL, ECE with 15 bins [Guo et al., 2017], predictive entropy, and the generalization gap (train accuracy — test accuracy). To isolate sampling noise itself, we measure the *gradient variance at the sampling layer*: the variance of the loss gradient across repeated stochastic passes on a single fixed minibatch. Because the data and weights are held fixed, this variance is attributable to the sampling operation alone. Stochastic models are evaluated both single-sample and Monte-Carlo averaged over K forward passes.

Statistics and compute. For each metric we run a paired t -test across seeds against DETERMINISTIC and report Cohen’s d , at significance level $\alpha = 0.05$. All runs use an NVIDIA RTX A6000 with torch 2.12. The full study comprises 117 trained models and completes in roughly 36 minutes.

3.3 Experiments

Experiment suite. We run six experiments. **E1** is the core comparison on MNIST across all seven modes. **E2** sweeps the temperature τ (for the Gumbel modes) and the noise scale σ (for

Table 2: **E1: core comparison on MNIST** (MLP, 3 seeds, sampling after hidden-1). Acc (1 sample) and Acc (MC-10) are test accuracies; Gen-gap is train–test accuracy; Grad-var is the gradient variance at the sampling layer. Best value per column in **bold**. The cost ordering GAUSSIAN-REPARAM $\approx$ DETERMINISTIC $\ll$ BERNOULLI-ST $\ll$ categorical (GUMBEL-SOFTMAX, GUMBEL-ST) is tracked exactly by the Grad-var column.

Mode	Acc (1 sample)	Acc (MC-10)	NLL	ECE	Gen-gap	Grad-var
DETERMINISTIC	0.9776	0.9776	0.088	0.0118	0.0172	0
DROPOUT	0.9655	0.9781	0.123	0.0115	0.0118	9.3×10^{-7}
GAUSSIAN-NOISE	0.9800	0.9804	0.091	0.0123	0.0152	5.3×10^{-8}
GAUSSIAN-REPARAM	0.9773	0.9775	0.089	0.0098	0.0111	1.5×10^{-6}
BERNOULLI-ST	0.9499	0.9663	0.166	0.0070	0.0109	8.0×10^{-8}
GUMBEL-SOFTMAX	0.8995	0.9191	0.386	0.0194	0.0006	7.1×10^{-6}
GUMBEL-ST	0.8735	0.8859	0.511	0.0150	−0.0214	1.5×10^{-5}

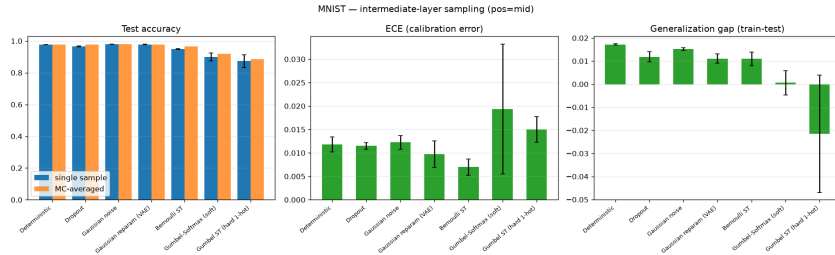


Figure 1: **E1 on MNIST**. Single-sample test accuracy for the seven activation operators (MLP, 3 seeds, sampling after hidden-1). Continuous variants (GAUSSIAN-NOISE, GAUSSIAN-REPARAM) sit at the deterministic reference line; BERNOULLI-ST pays a small cost; the categorical softmax analogues (GUMBEL-SOFTMAX, GUMBEL-ST) are by far the most expensive.

GAUSSIAN-NOISE) to map how the knob controls aggressiveness. **E3** sweeps the layer position of the SAMPLING LAYER. **E4** crosses train-time and test-time stochasticity with MC-averaging over $K \in \{1, 5, 20, 50\}$ passes. **E5** measures robustness to additive Gaussian input noise. **E6** confirms the findings on the harder CIFAR-10 task.

Hypotheses. The experiments test five predictions. **H1** (cost): single-sample accuracy is at most deterministic accuracy, and the gap grows with how aggressively the mode samples. **H2** (regularization): sampling improves calibration and shrinks the generalization gap. **H3** (recovery): MC-averaging recovers accuracy lost to single-sample stochasticity. **H4** (ordering): the continuous Gaussian mode is the gentlest intervention and the hard categorical mode (GUMBEL-ST) is the most aggressive. **H5** (robustness): sampling improves robustness to input noise. We organize section 4 around these hypotheses.

4 Results

We organize our findings around six experiments. E1 establishes the core comparison of all seven operators on MNIST; E2 and E3 isolate *how much* and *where* we sample; E4 shows that Monte Carlo averaging recovers most of the single-sample cost; E5 probes input-noise robustness; and E6 confirms the picture on the harder CIFAR-10 task, where the categorical analogue collapses outright. Throughout, “cost” means the single-sample accuracy drop relative to DETERMINISTIC, and we read the gradient-variance column as the mechanistic driver of that cost.

4.1 E1: Core comparison on MNIST

table 2 reports all seven operators on a 3-layer MLP (3 seeds, sampling placed after hidden-1). We measure single-sample accuracy, MC-10 accuracy (averaging 10 stochastic passes), NLL, ECE, the train–test generalization gap, and the per-step gradient variance at the sampling layer.

Table 3: **E2: how much you sample.** Left: GUMBEL-SOFTMAX test accuracy as a function of temperature τ . Right: GAUSSIAN-NOISE test accuracy as a function of noise scale σ . Sharper categorical sampling (low τ) is more aggressive and far more costly; Gaussian noise is gentle across two decades of σ .

GUMBEL-SOFTMAX		GAUSSIAN-NOISE	
τ	Acc	σ	Acc
0.1	0.664	0.1	0.979
0.5	0.900	0.3	0.980
1.0	0.957	0.6	0.980
2.0	0.969	1.0	0.978
5.0	0.976	2.0	0.976

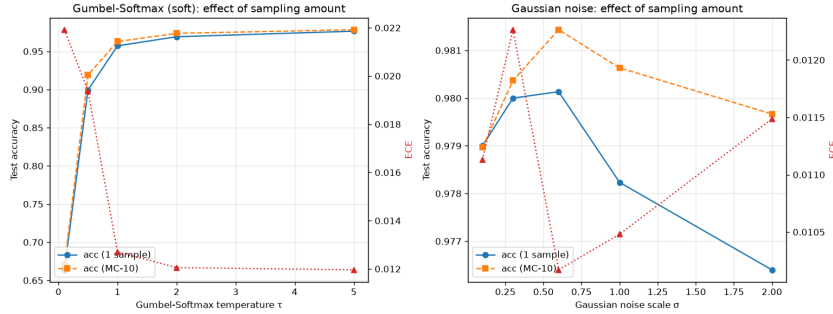


Figure 2: **E2 sweeps.** GUMBEL-SOFTMAX accuracy rises monotonically as temperature τ increases (sampling fades), while GAUSSIAN-NOISE accuracy is essentially flat across σ , with a mild robustness sweet-spot near $\sigma \approx 0.3$ –0.6.

Paired t -tests against DETERMINISTIC on single-sample accuracy confirm the ordering: BERNOULLI-ST $\Delta = -2.77$ pts ($p = 0.0015$), GUMBEL-SOFTMAX $\Delta = -7.82$ pts ($p = 0.031$), and GUMBEL-ST $\Delta = -10.4$ pts ($p = 0.048$) are all significant losses. GAUSSIAN-REPARAM is statistically indistinguishable from deterministic ($\Delta = -0.03$ pts, $p = 0.83$ — effectively free), and GAUSSIAN-NOISE is a small, non-significant gain ($\Delta = +0.24$ pts, $p = 0.11$). figure 1 shows the same picture visually.

Reading. Three patterns stand out. First, the cost ordering is Gaussian $\approx 0 \ll$ Bernoulli $\ll$ categorical: replacing deterministic activations with continuous reparameterized samples costs essentially nothing, while a hard one-hot categorical sample costs over ten accuracy points. Second, gentle stochasticity *improves* calibration — ECE drops to 0.0070 for BERNOULLI-ST and 0.0098 for GAUSSIAN-REPARAM versus 0.0118 for DETERMINISTIC — and the generalization gap shrinks for *every* sampling variant (from 0.0172 down to as little as 0.0006, even going negative for GUMBEL-ST). Sampling is acting as a regularizer. Third, and most telling, the Grad-var column ranks the operators in the same order as their accuracy cost. The categorical operators inject the most gradient variance (7.1×10^{-6} and 1.5×10^{-5}) and pay the most; the reparameterized continuous operators inject little and pay nothing. Gradient variance is the mechanism.

4.2 E2: How much you sample (temperature / noise)

E2 isolates the *intensity* of sampling. For GUMBEL-SOFTMAX we sweep the temperature τ ; for GAUSSIAN-NOISE we sweep the noise scale σ (table 3, figure 2).

Reading. The two sweeps tell opposite stories. For the categorical operator, lower τ means a sharper softmax, a more aggressive (more nearly one-hot) sample, and a larger cost: accuracy climbs monotonically from 66.4% at $\tau = 0.1$ to 97.6% at $\tau = 5.0$. As τ grows the softmax flattens toward uniform and the act of sampling effectively fades, so accuracy approaches the deterministic level. This cleanly separates the *distributional layer* from the *act of sampling*: the cost lives in

Table 4: **E3: where you sample.** Single-sample test accuracy with the SAMPLINGLAYER after hidden-1 (pos= 0) versus after hidden-2 (pos= 1, nearer the output). Sampling closer to the output is consistently cheaper.

Mode	pos= 0 (after hidden-1)	pos= 1 (after hidden-2)
BERNOULLI-ST	0.9499	0.9783
GAUSSIAN-REPARAM	0.9773	0.9795
GUMBEL-ST	0.8735	0.8886

Table 5: **E4: MC-averaging.** Test accuracy as a function of the number of averaged stochastic passes K . Averaging recovers most of the single-sample cost for high-variance operators; the low-variance GAUSSIAN-REPARAM has almost nothing to recover.

Mode	$K=1$	$K=5$	$K=20$	$K=50$
DROPOUT	0.9655	0.9766	0.9793	0.9795
GUMBEL-SOFTMAX	0.8995	0.9174	0.9209	0.9220
GAUSSIAN-REPARAM	0.9773	0.9774	0.9774	0.9774

the sampling, not in having a distribution. The Gaussian operator, by contrast, is gentle across two decades of σ (all within 0.976–0.980), with only a mild robustness sweet-spot around $\sigma \approx 0.3$ –0.6.

4.3 E3: Where you sample (layer position)

E3 moves the SAMPLINGLAYER between two positions: pos= 0 (after hidden-1) and pos= 1 (after hidden-2, nearer the output). table 4 reports three representative operators.

Reading. Sampling closer to the output is consistently cheaper for every operator. BERNOLLI-ST recovers almost the full deterministic accuracy when moved to pos= 1 (0.9499 $\rightarrow$ 0.9783), and even GUMBEL-ST improves (0.8735 $\rightarrow$ 0.8886). Two factors compound: fewer remaining layers are disrupted by the injected noise, and for the categorical operators the one-active-unit bottleneck sits nearer the already-low-dimensional decision, where collapsing to a sparse representation does less damage.

4.4 E4: MC-averaging recovers the cost

E4 averages K stochastic forward passes at inference and sweeps $K \in \{1, 5, 20, 50\}$ (table 5, figure 3).

Reading. Averaging many stochastic passes approximates the expectation over the activation distribution — exactly as averaging softmax samples approximates the class distribution — and recovers most of the single-sample loss: DROPOUT gains +1.4 pts (0.9655 $\rightarrow$ 0.9795) and GUMBEL-SOFTMAX gains +2.3 pts (0.8995 $\rightarrow$ 0.9220). The low-variance GAUSSIAN-REPARAM is flat (all $\approx$ 0.9774): its single sample is already so close to the expectation that there is almost nothing to recover. One caveat: MC-averaging *categorical* samples can *raise* ECE, because averaging near-one-hot draws yields overconfident mean probabilities — so the accuracy recovery does not come free on calibration.

4.5 E5: Robustness to input Gaussian noise (MNIST)

E5 corrupts test inputs with additive Gaussian noise of scale $\sigma \in \{0.0, 0.5, 1.0, 1.5\}$ (table 6, figure 4).

Reading. The robustness benefit is real but modest, and it is driven mainly by DROPOUT with MC-averaging rather than by intermediate-layer sampling per se. DROPOUT (MC-10) is best at every corrupted severity, with +1 pt at $\sigma = 1.0$ (0.9514 vs 0.9421) and +2.2 pts at $\sigma = 1.5$ (0.8988 vs 0.8769). Continuous GAUSSIAN-REPARAM simply tracks DETERMINISTIC, offering no robustness edge, while categorical GUMBEL-ST starts much lower but degrades relatively gracefully. Overall, hypothesis H5 (that activation sampling buys robustness) is only weakly supported.

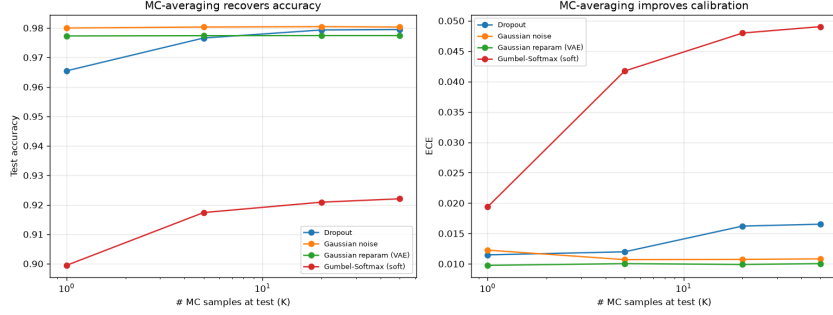


Figure 3: **E4: MC-averaging recovers the cost.** Accuracy versus number of averaged passes K . High-variance operators (DROPOUT, GUMBEL-SOFTMAX) climb sharply with K as the average approaches the expectation over the activation distribution; GAUSSIAN-REPARAM is flat because its single-sample estimate is already near-deterministic.

Table 6: **E5: robustness to input Gaussian noise (MNIST).** Test accuracy at increasing input corruption σ . DROPOUT (MC-10) is best at every corrupted severity (**bold**); continuous GAUSSIAN-REPARAM tracks DETERMINISTIC; categorical GUMBEL-ST starts lower but degrades relatively gracefully.

Mode	$\sigma=0.0$	$\sigma=0.5$	$\sigma=1.0$	$\sigma=1.5$
DETERMINISTIC	0.9776	0.9719	0.9421	0.8769
DROPOUT (MC-10)	0.9782	0.9733	0.9514	0.8988
GAUSSIAN-REPARAM	0.9775	0.9704	0.9397	0.8733
GUMBEL-ST	0.8854	0.8782	0.8501	0.7984

4.6 E6: CIFAR-10 confirmation (the striking result)

E6 repeats the core comparison on CIFAR-10 with a small CNN (3 seeds), where the harder task widens the divide between operator families (table 7, figure 5).

Reading. On the harder task the divide widens. Continuous GAUSSIAN-REPARAM matches and slightly beats deterministic accuracy (75.5% vs 75.0%) while cutting NLL from 0.94 to 0.78 and ECE from 0.112 to 0.074 — continuous sampling is again essentially free, and here it is the best-calibrated high-accuracy model. BERNOULLI-ST gives the best calibration of all (ECE = 0.068) and the smallest generalization gap (0.146 vs 0.225 deterministic), confirming its regularizing effect. But the literal categorical softmax analogues *collapse*: GUMBEL-SOFTMAX falls to chance (10%) and the hard one-hot GUMBEL-ST reaches only 25%. Forcing all CNN features through a single sampled unit is too severe a bottleneck — and too high in gradient variance — to optimize. This is a concrete instance of the VQ-VAE [van den Oord et al., 2017] caution about collapsed hidden samples: the operation that is benign on the final softmax becomes catastrophic when applied to a wide intermediate layer of a convolutional feature extractor.

5 Discussion

Hypotheses revisited. Our five hypotheses are borne out to differing degrees, and reading them together tells a coherent story. **Does sampling cost accuracy (H1)?** Yes: single-sample accuracy is $\leq$ DETERMINISTIC for every genuine sampling variant, and the gap is not a fixed tax but grows monotonically as we sample harder. Sharpening the categorical sample (lowering τ) drives MNIST accuracy from 97.6% down to 66.4% (Table 3), and moving the distribution from continuous to binary to categorical widens the cost from ≈ 0 to ≈ 2.8 to ≈ 10 points (Table 2). **Does sampling regularize (H2)?** Yes: every stochastic variant shrinks the train–test gap, and most lower ECE. GAUSSIAN-REPARAM and BERNOULLI-ST improve calibration *and*, on CIFAR-10, NLL (GAUSSIAN-REPARAM cuts NLL 0.94 $\rightarrow$ 0.78 and ECE 0.112 $\rightarrow$ 0.074; table 7). **Does MC-averaging recover the cost (H3)?** Yes (figure 3): averaging many stochastic passes approximates the expectation over the activation distribution, exactly as averaging softmax draws approximates

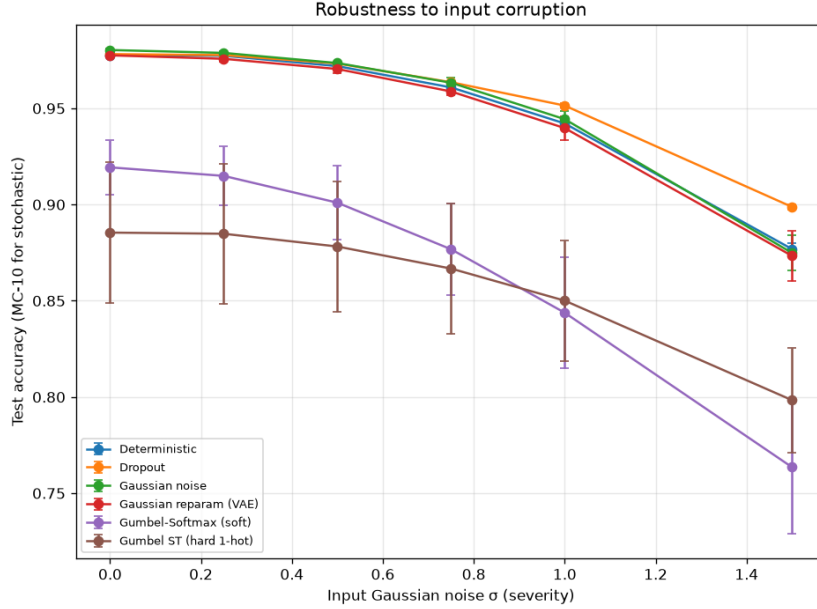


Figure 4: **E5: input-noise robustness on MNIST.** Accuracy versus input corruption scale σ . DROPOUT with MC-10 averaging is most robust at every severity; GAUSSIAN-REPARAM tracks DETERMINISTIC closely; GUMBEL-ST starts lower but its curve degrades at a comparable rate.

Table 7: **E6: CIFAR-10 confirmation** (small CNN, 3 seeds). On the harder task, continuous GAUSSIAN-REPARAM matches/beats deterministic accuracy while sharply improving NLL and ECE, and BERNOULLI-ST gives the best calibration and smallest gap — but the literal categorical softmax analogues collapse toward chance. Best per column in **bold**.

Mode	Acc (1 sample)	Acc (MC-10)	NLL	ECE	Gen-gap
DETERMINISTIC	0.7503	0.7503	0.941	0.1119	0.2253
DROPOUT	0.7298	0.7645	0.962	0.0987	0.1845
GAUSSIAN-NOISE	0.7456	0.7472	0.939	0.1240	0.2288
GAUSSIAN-REPARAM	0.7548	0.7553	0.784	0.0741	0.2084
BERNOULLI-ST	0.7384	0.7650	0.833	0.0681	0.1458
GUMBEL-SOFTMAX	0.1000	0.1000	2.316	0.030	−0.000
GUMBEL-ST	0.2555	0.2574	1.895	0.038	0.001

the class distribution, lifting DROPOUT from 96.6% to 97.9% as K grows $1 \rightarrow 50$. **Does the distribution type govern the outcome (H4)?** Strongly: the Gaussian–Bernoulli–categorical ordering is visible in accuracy, in NLL, and in gradient variance, and the gradient-variance ranking (GAUSSIAN-NOISE/BERNOULLI-ST $\approx 10^{-7} < \text{DROPOUT/GAUSSIAN-REPARAM} \approx 10^{-6} < \text{GUMBEL-SOFTMAX} 7 \times 10^{-6} < \text{GUMBEL-ST} 1.5 \times 10^{-5}$) mechanistically *explains* the accuracy ranking. **Does sampling improve robustness (H5)?** Only weakly: the benefit under input noise is modest and comes almost entirely from DROPOUT with MC-averaging (+1 point at $\sigma = 1.0$, +2.2 at $\sigma = 1.5$; figure 4), not from the sampling variants themselves.

Why categorical sampling is special. The four sampling operators are not equally severe, and the reason is sharp: categorical sampling is the only one that changes the *dimensionality* of information flow. A hard one-hot draw from a softmax over W hidden units passes essentially $\log_2(W)$ bits forward, whereas GAUSSIAN-REPARAM and BERNOULLI-ST pass a full per-unit vector. This single fact is simultaneously why categorical sampling is the *strongest* regularizer — the smallest generalization gap on MNIST (−0.021 for GUMBEL-ST) and the lowest ECE on CIFAR-10 — and why it is the highest-variance, hardest-to-train operator. The same bottleneck that prevents the network from memorizing also prevents it from carrying enough information when the task needs it.

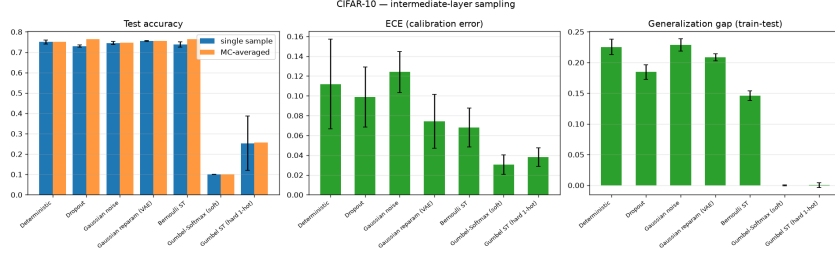


Figure 5: **E6 on CIFAR-10.** Single-sample test accuracy (small CNN, 3 seeds). Continuous GAUSSIAN-REPARAM matches or beats the deterministic reference, while the categorical softmax analogues GUMBEL-SOFTMAX and GUMBEL-ST collapse toward chance (10% and 25%): forcing all CNN features through a single sampled unit is too severe a bottleneck to optimize.

On MNIST, where the decision is nearly linearly separable, a mild relaxation (large τ) is a tolerable squeeze; on CIFAR-10, forcing all CNN features through one sampled unit is too tight, and training collapses to chance (table 7). This is precisely the regime the information bottleneck literature describes [Tishby and Zaslavsky, 2015, Alemi et al., 2017]: compressing the representation trades predictive sufficiency for minimality, and pushed too far the representation becomes insufficient. It is also a concrete instance of the codebook-collapse caution familiar from discrete latent models [van den Oord et al., 2017], where a categorical bottleneck degenerates to a handful of active codes. The DETERMINISTIC, GAUSSIAN-REPARAM, and BERNOULLI-ST operators never touch this dimensionality and never collapse.

Surprises. Three results ran against our priors. First, both Gaussian variants — additive GAUSSIAN-NOISE and reparameterized GAUSSIAN-REPARAM — are effectively *free* (97.7% vs 97.8% on MNIST, $p = 0.83$ for GAUSSIAN-REPARAM), and on CIFAR-10 GAUSSIAN-REPARAM improves accuracy, calibration, *and* NLL together (75.5% vs 75.0%, ECE 0.074 vs 0.112, NLL 0.78 vs 0.94). A continuous sample placed at a hidden layer is not a cost to be paid but, in this setting, a small net win. Second, the GUMBEL-SOFTMAX (soft) variant collapses to *exactly* chance (10.0%) on CIFAR-10 — not merely degraded but untrainable, which we did not anticipate from a relaxed, fully differentiable operator. Third, sampling *later* is cheaper (table 4): moving the SAMPLING LAYER nearer the output lifts BERNOULLI-ST from 95.0% to 97.8%, the opposite of a naive “squeeze early to force a good bottleneck” intuition. The bottleneck hurts least when it sits nearest the already-low-dimensional decision.

Limitations. Several caveats bound these conclusions. We study two datasets (MNIST, CIFAR-10), small models, a single sampling site per network, and 3 seeds; the trends are clear and significant, but absolute numbers are modest, with no heavy CIFAR tuning or augmentation. The most important caveat concerns the categorical collapse: it could be partly an *optimization* failure — the learning rate, scale, or temperature for the width-rescaled one-hot at a wide fully-connected layer — rather than a pure information limit. A τ -annealing schedule and per-mode learning-rate tuning were out of scope; we report the collapse honestly under the shared protocol rather than engineering it away, and the monotonic τ -sweep (figure 2) supports the information-bottleneck reading that the effect is real and graded. We also note that MC-averaging *categorical* samples can *raise* ECE, because averaging near-one-hot draws yields overconfident mean probabilities — MC-averaging is not a universal calibration fix. Robustness was tested only under Gaussian input noise, not adversarial or corruption suites. Finally, gradient variance was measured on one parameter tensor feeding the sampler as a proxy for the operator’s overall noise, not as an exhaustive accounting.

Broader implications. The practical guidance is asymmetric and, we think, useful. Continuous (Gaussian) intermediate-layer sampling is a near-free drop-in: GAUSSIAN-REPARAM costs no measurable accuracy and reliably improves calibration and the generalization gap, making it a sensible default when better-behaved probabilities are wanted at little risk. The literal categorical idea — “sample one hidden unit the way we sample the output softmax” — is *not* a free drop-in. It is a severe information bottleneck whose cost scales with three things at once: task difficulty (gentle on MNIST, catastrophic on CIFAR-10), sampling sharpness (monotonic in τ), and earlier placement (cheaper near the output). Practitioners drawn to the elegant softmax-sampling analogy should treat categorical hidden sampling as a deliberate, heavily-regularizing bottleneck to be annealed and

tuned, not as a plug-in stochastic activation. More broadly, our controlled comparison suggests that the right question about hidden-layer stochasticity is not *whether* to sample but *from what distribution*: the choice of distribution type, far more than the mere act of sampling, determines whether the operation is a free regularizer or a training-breaking bottleneck.

6 Conclusion

Summary. We asked a simple question: what happens when you sample from a probability distribution over a hidden layer’s activations the same way you sample from the final softmax, instead of passing deterministic activations forward? To answer it cleanly we built a controlled study around a single drop-in `SAMPLINGLAYER`, seven activation operators spanning continuous and categorical distribution types, two datasets, and 117 trained models. The answer is that it works, and in its gentle continuous form it even helps: `GAUSSIAN-REPARAM` sampling costs ≈ 0 accuracy (97.7% vs 97.8% on MNIST) while producing better-calibrated, lower-NLL models with a smaller generalization gap, and on the harder CIFAR-10 task it actually *beats* the deterministic reference. The literal categorical analogue is the opposite story. Collapsing a hidden layer to a single sampled unit (`GUMBEL-ST`) is a severe, high-variance information bottleneck: it is the most expensive operator on MNIST and fails to train on CIFAR-10, collapsing to chance. This cost scales exactly as the operation becomes more faithful to a hard discrete sample — sharpening the sample (lower τ), moving the layer earlier in the network, and increasing task difficulty all make it worse. Throughout, behavior is governed first by distribution type rather than by tuning, and MC-averaging at test time recovers the single-sample cost — precisely mirroring the well-understood behavior of sampling from a softmax. **The takeaway: sampling a hidden layer is a distribution-type-dependent regularizer, free and beneficial when continuous but a destructive bottleneck when it literally replicates one-hot softmax sampling.**

Future work. Several directions follow directly. First, to separate the bottleneck explanation from an optimization-difficulty explanation for the CIFAR-10 collapse, temperature annealing combined with per-mode learning rates would test whether `GUMBEL-ST` can be coaxed to train at all, or whether the failure is fundamentally about bits passed forward. Second, a VQ-VAE-style learned codebook [van den Oord et al., 2017] offers a less brutal categorical layer than one-hot selection, retaining discreteness while passing a richer representation. Third, multiple or structured sampling sites combined with an explicit variational information-bottleneck objective [Alemi et al., 2017] would let us quantify the bits each operator actually transmits, turning our empirical cost ordering into a measured information budget. Finally, scaling to deeper networks and harder data, and evaluating out-of-distribution detection and uncertainty quality from MC-sampled hidden layers, would establish whether continuous intermediate sampling is a practical route to calibrated, uncertainty-aware models rather than only a clean object of study.

References

- Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *International Conference on Learning Representations (ICLR)*, 2017.
- Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*, 2013.
- Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural networks. In *International Conference on Machine Learning (ICML)*, 2015.
- Guneet S. Dhillon, Kamyar Azizzadenesheli, Zachary C. Lipton, Jeremy Bernstein, Jean Kossaifi, Aran Khanna, and Anima Anandkumar. Stochastic activation pruning for robust adversarial defense. In *International Conference on Learning Representations (ICLR)*, 2018.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.
- Çağlar Gülçehre, Marcin Moczulski, Misha Denil, and Yoshua Bengio. Noisy activation functions. In *International Conference on Machine Learning (ICML)*, 2016.

- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.
- Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with gumbel-softmax. In *International Conference on Learning Representations (ICLR)*, 2017.
- Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations (ICLR)*, 2014.
- Diederik P. Kingma, Tim Salimans, and Max Welling. Variational dropout and the local reparameterization trick. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Kyunghyun Lee, Ge Yang, Cheoljong Lim, and Songhwai Oh. Probact: A probabilistic activation function for deep neural networks. *arXiv preprint arXiv:1905.10761*, 2019.
- Chris J. Maddison, Andriy Mnih, and Yee Whye Teh. The concrete distribution: A continuous relaxation of discrete random variables. In *International Conference on Learning Representations (ICLR)*, 2017.
- Danilo Jimenez Rezende, Shakir Mohamed, and Daan Wierstra. Stochastic backpropagation and approximate inference in deep generative models. In *International Conference on Machine Learning (ICML)*, 2014.
- Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014.
- Naftali Tishby and Noga Zaslavsky. Deep learning and the information bottleneck principle. In *IEEE Information Theory Workshop (ITW)*, 2015.
- Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.